

Puzzles

COMPLETE LEARNING GUIDE

What You'll Learn

Logical puzzles — ranking, distribution, floor-based, and box/grid arrangements — test systematic deduction under a strict set of constraints rather than raw calculation, and the biggest score difference between candidates comes from HOW they organize the clues, not from any special mathematical ability. Attempting to hold every clue in your head at once is what causes candidates to freeze; the fix is external representation — a grid, a number line, or a simple table — filled in step by step, applying the most restrictive (most specific) clue first. This sub-module walks through the core puzzle families with fully solved examples: a preference-matching grid, a linear ranking puzzle, a distribution puzzle with algebraic constraints, a floor-based puzzle, a box/position puzzle, conditional (if-then) chain deduction, and a time-management approach for puzzle sets, which are usually worth several questions each and reward getting the setup right once.

1. Grid Method for Preference-Matching Puzzles

When each of several people is matched to exactly one item from a set (fruit, subject, sport), draw a grid of people vs items and mark each clue as a confirmed match, an elimination, or leave blank — apply the most specific clues (direct matches) before the exclusionary ones, since a single confirmed match often forces several others by elimination.

- Aisha, Bala, Chetan, Divya each like exactly one of mango, apple, grape, banana. Clues: Aisha doesn't like mango or banana; Bala likes grape; Chetan doesn't like apple; Divya likes mango. Since Divya=mango, Aisha (not mango, not banana, grape taken by Bala) must like apple. Chetan (not apple, apple already taken anyway) gets the only fruit left: banana. Final: Aisha–apple, Bala–grape, Chetan–banana, Divya–mango.

2. Linear Ranking Puzzles

Ranking puzzles (1st to last place) are solved by first placing any absolutely-fixed clues (e.g., 'X is ranked 2nd'), then working through relative clues ('immediately above/below') against the remaining open slots.

- Five students P, Q, R, S, T are ranked 1 (best) to 5 (worst). Clues: R is ranked 2nd; S is ranked last (5th); Q is immediately above T. Remaining open ranks for P, Q, T are {1, 3, 4}. The only consecutive pair within {1,3,4} is (3,4), so Q=3, T=4, leaving P=1. Final order: P, R, Q, T, S.

3. Distribution Puzzles With Algebraic Constraints

Distribution puzzles (items shared among people with relative-quantity clues) are often fastest solved by assigning a variable to the smallest unknown and expressing every other quantity in terms of it, then solving the resulting equation from the total.

- 20 chocolates are split among P, Q, R such that P gets 4 more than Q, and R gets twice as many as P. Let $Q = x$. Then $P = x+4$, $R = 2(x+4)$. Sum: $x + (x+4) + 2(x+4) = 20$!' $4x + 12 = 20$!' $x = 2$. So $Q=2$, $P=6$, $R=12$ (check: $2+6+12=20$ ').

4. Floor-Based Puzzles

Floor puzzles (people living on numbered floors) work like ranking puzzles but with the added twist of 'above/below' language mapping directly to higher/lower floor numbers — fix the anchor clues first (specific floor, topmost/bottommost), then work outward.

- Six floors (1=bottom to 6=top), six people A–F. Clues: A is on floor 3; C is on the topmost floor (6); D is immediately below C (floor 5); B is one floor above A (floor 4). That leaves floors 1 and 2 for E and F — genuinely undetermined without a further clue, which is common: not every puzzle resolves every person, and a well-set question only asks about what CAN be determined.

5. Box and Position Puzzles

Box or position-in-a-row puzzles (cars in parking slots, boxes on a shelf) follow the same discipline as ranking puzzles: place absolute-position clues first (an exact slot number or 'at an end'), then use relative clues to fill in what remains.

- 5 positions in a row (1 to 5, left to right): Red is at position 3. Blue is immediately right of Red (so $\text{Blue}=4$). Green is at an end (1 or 5). Black is not at an end. White is at position 1. Since $\text{White}=1$, Green (must be an end, and 1 is taken) must be 5. The only non-end slot left, 2, goes to Black. Final: White, Black, Red, Blue, Green.

6. Chain Deduction With Conditional (If-Then) Statements

Some puzzles present a chain of conditional statements rather than positions — the technique is to trace the chain forward from whatever fact is confirmed true, applying each 'if-then' link in sequence until you reach the final conclusion.

- 'If it rains, the match is postponed. If the match is postponed, the team practices indoors. It rained today.' Chain: rain !' postponed !' indoor practice. Conclusion: the team practiced indoors today.

7. Time Management for Puzzle Sets

Puzzle sets are usually worth 3-5 questions tied to one setup, so the setup time is amortized across all of them — investing an extra 60-90 seconds to build a correct, complete grid up front is almost always worth it, versus rushing the setup and having to redo work when a contradiction appears midway.

- If a puzzle takes 3 minutes to set up correctly but then yields 5 quick questions at under 30 seconds each, total time is about 5.5 minutes for 5 marks — far better than guessing at an incomplete or contradictory setup and losing several of those marks.

Common Pitfalls

- Trying to solve a multi-person puzzle mentally without drawing a grid or table, leading to lost track of earlier deductions.
- Applying clues in the order they're written instead of applying the most restrictive (absolute-position) clues first.
- Misreading 'immediately above/below' as simply 'above/below' (any distance), which allows too many possibilities and causes contradictions later.
- Assuming every person/position in a puzzle must be fully determinable — some puzzles genuinely leave 2 people interchangeable if no clue distinguishes them.
- Not double-checking a completed grid against every original clue before answering, missing a clue that was overlooked mid-solve.
- Spending too little time on setup and guessing at questions once a contradiction appears, instead of re-checking clue placement order.

Key Takeaways

- Always externalize a multi-constraint puzzle into a grid, table, or number line — never solve it purely in your head.
- Apply absolute/fixed clues (exact position, 'topmost', 'at an end') before relative clues ('immediately above', 'next to').
- 'Immediately above/below/left/right' means directly adjacent; plain 'above/below/left/right' allows any distance.
- Not every puzzle fully determines every element — some ambiguity between two people can be a legitimate, intended outcome.
- For distribution puzzles, assign a variable to the smallest unknown and express the rest algebraically in terms of it.
- Invest adequate setup time on puzzle sets — the cost is amortized across several linked questions.

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IndiaBix — Logical Reasoning Puzzles

<https://www.indiabix.com/logical-reasoning/puzzles/>

GeeksforGeeks — Logical Reasoning Questions and Answers

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